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(1) Research Program/Project/Study Title: Growth and Yield of Radish and Changes in Soil Properties Associated with the Application of Different Compost Forms

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Program or Project has component (if no, proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation

Municipality/City:	
Province:	
Region:	

(5) Classification

Research:

☐ Basic

☐ Applied
Promotion/Commercialization

Development:

☐ Pilot Testing

☐ Tech.

(6) Mode of Implementation

- ☐ Single Authorship/Proponent
☐ Collaborative Research (within campus)
☐ Collaborative Research (intercampus)
☐ Collaborative Research (external agency)

(7) Sector/Commodity

(8) Discipline/Field of Specialization

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, significant findings, and implications of the impact of the study.)

This study determined the growth and yield of radish and changes in soil properties associated with the application of different compost forms. The experiment was established in Randomized Complete Block Design (RCBD) and replicated thrice. The different compost forms are Vermicast, F2- Vermi tea, and F3 –Ordinary Compost. Applying vermicast on radish plant statistically gave the best growth performance in terms of width of the widest leaf.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words are already given keywords. This would also include process or methodology employed.)

Raphanus sativus, ordinary compost, vermicast, vermi tea, radish

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Radish is a member of the Cruciferae (mustard family). *Raphanus sativus* is a cool season annual (depending on when it is planted). Radishes have been cultivated for thousands of years in both China and the Mediterranean area. Radishes were a common food in Egypt before the building of the pyramids and is one of our most ancient cultivated plants (APA). Radishes eaten raw are rich in folate (folic acid), calcium, potassium, and dietary fiber. This vegetable is not very rich in Vitamin A of

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B-complex vitamins but is moderately high in Vitamin C. The sodium content is also moderately high. However, radishes are low in calories, fat, and cholesterol (Agrilife Extension, 2009).

World production of radish roots is estimated at 7 million t per year, about 2% of the total world production of vegetables (Schippers 2004). In China, Japan and Korea, three countries which are known consumer of radish in their cuisine, as well as in Yemen, radish ranks high in importance (Schippers 2004).

Vermicompost, an organic fertilizer, is now becoming an exciting approach and proven to be an effective means of improving soil structure, enhancing soil fertility, and increasing crop yield. Plant growth is highly stimulated by vermicompost; because it contains nutrients and other essential nutrients like phosphorus and potassium. Joshi et al. (2014) described vermicompost as an excellent soil amendment and a biocontrol agent through several studies, along with the reasons which make it the best organic fertilizer and more eco-friendly than chemical fertilizers. To evaluate the response of Radish to the application of various forms of compost, this study was conducted.

The study specifically aimed to:

1. Determine the growth and yield of radish as affected by compost applied in various forms;
2. Assess the changes in soil quality indices as affected by compost application in various forms; and
3. Evaluate the nutrient uptake of radish applied with compost in various forms.

(11) Materials and Methods or Methodology (This includes the study's materials, methods, duration & location. Materials and Methods are headings used in Natural Science research and methodology in Social Sciences. Presentation of materials may not necessarily be in a separate heading but may be incorporated into the specific methods used. Justify paragraphs with main headings in caps.)

The materials and equipment that were used in the study were the following: seeds of sweetcorn, bamboo, fertilizer, insecticide, fungicide, knapsack sprayer, plastic bags, straw lace, bolo, shovel, hammer, nails, digital weighing scale, pen, calculator, tape measure, caliper, data notebook, sacks, and tarpaulin.

Experimental Design. The study was composed of different compost forms: F1 – Vermicast, F2- Vermi tea, and F3 –Ordinary Compost. The study was laid out using the Randomized Complete Block Design with three replications.

Cultural Practices

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Land Preparation. Nine plots measuring 4 m x 4 m were prepared to accommodate the 3 treatments replicated. Each block was spaced 1 m while the distance between plots was 0.5 m. Border plots with a width of 1 m were provided. The area was prepared by plowing and harrowing twice with an interval of one week per operation before the construction of plots and paddies.

Securing the seeds. The radsih seeds were bought from a reputable agricultural supplier in the province.

Sowing of seeds. The radish seeds were sown in a seed box containing garden soil, compost, and sand in equal proportion. Seeds were sown in a prepared seed box by evenly spreading them in a finger-made furrow.

Pricking of seedlings. As soon as the seeds germinated, the overcrowded germinated seeds were transferred to the seedling trays to allow them more space to grow and avoid root intertwining. The seedlings were kept in the nursery until they produced three true leaves and a vigorous appearance before being transplanted into the field.

Hardening off seedlings. Seedlings were hardened as soon as they produced true leaves by gradually exposing the seedlings to open sunlight to avoid transplanting shock during transplanting. This was done by exposing the seedlings to open sunlight from early morning until 9:00 a.m. on the first week. The number of exposures to sunlight increased in the following weeks until they produced three or four true leaves and were ready for transplanting.

Transplanting the seedlings. The seedlings with three to four true leaves, vigorous and disease-free, were transplanted into the fields following the recommended planting distance. Transplanting was done late in the afternoon to minimize transplanting shock.

Care of seedlings. The transplanted seedlings were initially covered with banana bracts for four days during extreme sunlight to prevent them from wilting. Plant watering using sprinklers was done every morning and late in the afternoon.

Cultivation and weeding. The cultivation of space between rows and hills was done two weeks after transplanting. Shallow cultivation was practiced using a hand trowel to avoid mechanical injury of the roots. Hand weeding was done by pulling the weeds using a trowel as soon as the weeds appeared in the plots.

Watering. The plants were watered whenever necessary, especially when there was no rain.

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Application of different forms of compost. Vermicast and ordinary compost were applied to the designated experimental plots in the final stage of land preparation. The application rate was 200 grams per hill based on the recommended rate and analysis. For the vermi tea, the application will be made two weeks before transplanting and every week after that except during the flowering stage. The concentration used was 20 ml/liter of water. The application rate per plot was four liters, and the amount was increased by 1 liter every two weeks.

Pest and disease management and control. Cleanliness and sanitation were strictly implemented in the Project. Pest and disease monitoring were strictly done. In case of high pest and disease incidence, which will threaten the performance of plants, botanical, OHN, and organically formulated pesticides were used. Insect repellent plants were also planted around the experimental areas. The number and kinds of pests that may occur were recorded.

Harvesting/Priming. Radsih was harvested when it reached the marketable stage. The manifestation of the maturity indices is gauged at this stage. For the radish, harvest was done at 30 to 45 days after transplanting.

Data to be Gathered. The following data were gathered: plant height, number of leaves, width of widest leaf, diameter of tuber, length of tuber, biomass, weight of marketable part, weight of non-marketable part, total plot yield, total marketable yield, total non-marketable yield.

(12) Results and Discussions (This part contains the presentation of results (data) concerning the table and figure, to be followed by discussion, with support of reviewed literature. The sequence shall be based on the objectives of the study. Tables are formatted without vertical lines. Horizontal lines are set at the top and bottom portions, below table headings and significant subdivisions. Justify paragraphs with main headings in caps.)

Growth Parameters

Plant Height

Result shows (Table 1.0), that the radish applied F₁ - Vermicast obtained the tallest plant with a mean of 31.01 cm. This was followed by F₃ - Ordinary Compost and F₂ - Vermitea with the mean of 29.30 cm and 28.51 cm, respectively.

However, analyzed through Analysis of Variance, the plant height of radish was not affected by the application of different compost forms.

Number of Leaves

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Result (Table 1.0) shows that the radish applied with F₁ - Vermicast obtained the most number of leaves with the mean of 16.8 leaves. This was followed by F₂ - Vermitea with the mean of 16.23 number of leaves, and F₃ - Ordinary Compost with the mean of 14.7, which was the lowest among treatments.

However, analysis shows that the number of leaves was not affected by the application of different compost forms.

Width of Widest Leaf

Result shows that radish applied with F₁ - Vermicast, obtained the widest leaf with the mean of 10.82 cm, followed by F₂ - Vermitea and F₃ - Ordinary Compost with the means of 9.95 cm and 9.38 cm, respectively.

Furthermore, analysis revealed that the width of widest leaf of radish was significantly affected by the different compost forms. The highest mean was obtained from F₁ - Vermicast, and the lowest mean was obtained from F₃ - Ordinary Compost but was comparable to F₂ - Vermitea.

Table 1.0 Means of different growth parameters applied with different compost form.

Different Compost	Plant Height (cm)(ns) (CV =8.57%)	Number of Leaves (ns) (CV =7.67%)	Width of Widest Leaf (cm)(**) (CV =3.03%)	Biomass (g)(ns) (CV=21.30%)
F ₁ - Vermicast	31.01	16.8	10.82a	363.67
F ₂ - Vermitea	28.51	16.23	9.95b	282.67
F ₃ - Ordinary Compost	29.3	14.7	9.38b	226.00

Yield Parameters

Diameter of Tuber

Result shows that the radish applied with F₁ - Vermicast, obtained the biggest diameter with the mean of 4.31 cm. This was followed by F₂ - Vermitea with the mean of 3.94 cm and F₃ - Ordinary Compost with the mean of 3.70 cm.

However, results of the analysis revealed the diameter of tuber from the harvested radish was not affected by the application of different compost forms.

Length of Tuber

The results revealed that the radish applied with F₁ - Vermicast obtained the longest tuber among samples with the mean of 23.73 cm. This was followed by F₂ -

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Vermitea and F₃ - Ordinary Compost with the mean of 22.84 cm and 20.06 cm, respectively.

However, analysis revealed that the length of tuber was not affected by the application of different compost forms.

Biomass

Results revealed that the radish applied with F₁ - Vermicast obtained the most heaviest biomass with the mean of 363.67 g. This was followed by F₂ - Vermitea with the mean of 282.67 g and F₃ - Ordinary Compost with the mean of 226.00 g.

Analysis revealed that the biomass of radish was not affected by the application of different compost forms.

Weight of Marketable Part

Result revealed that the radish applied with F₁ - Vermicast obtained the heaviest marketable part with the mean of 257.67 g, followed by F₂ - Vermitea and F₃ - Ordinary Compost with the mean of 200.83 g and 157.00 g, respectively.

However, analysis revealed that the weight of marketable of radish was not significantly affected by the application of different compost form.

Weight of Non-marketable Part

Result revealed that the radish applied with F₁ - Vermicast obtained the heaviest non-marketable part, with the mean of 102.50 g. A derivative of vermicast, F₂ - Vermitea followed with the mean of 75.50 g, and the lowest weight was obtained by F₃ - Ordinary Compost with the mean of 68.33 g.

However, as anlysis revealed, the weight of non-marketable part was not affected by the application of different compsof forms.

Total Plot Yield

Result revealed that the radish applied with F₁ - Vermicast obtained the heaviest total plot yield with the mean of 19256.67 g. Followed by F₂ - Vermitea with the mean of 15281.67 g, which then the lightest among total plot yield was obtained by F₃ - Oradinary Compost with the mean of 13765.00 g.

However, analysis reveled the hthe total plot yield of radish was not affected by the application of different compost form.

Total Marketable Yield

Result revealed that the heaviest total marketable yield of radish was obtained from F₁ - Vermicast with the mean of 3548.33 g. This was followed by F₂ - Vermitea with the mean of 2481.67 g, and F₃ - Ordinary compost with the mean of 2065.00 g.

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However, analysis revealed that the total marketable yield of radish was no affected by the different compost form.

Totat Non-marketable Yield

Result revealed that the heaviest total non-marketaboe yield of radish was obtained from F₁ - Vermicast with the mean of 3548.33 g, this was followed by F₂ - Vermitea and F₃ - Ordinary compost, with thea mean of 2481.67 g and 2065.00 g, respectively.

However, analysis revealed that the total non-marketable yield of radish was no affected by the different compsof form.

Table 2.0 Means of different yield parameters applied with different compost form.

Different Compost	Diameter of tuber (cm) (ns)(CV = 11.85%)	Length of tuber (cm)(ns) (CV = 7.22%)	Weight of marketable part (g)(ns) (CV=23.00%)	Weight of non-marketable part (g)(ns) (CV=26.33)
F1 - Vermicast	4.31	23.73	257.67	102.5
F2 - Vermitea	3.94	22.84	200.83	75.5
F3 - Ordinary Compost	3.7	20.06	157	68.33

Plot yield (g)(ns) (CV=16.33%)	Marketable yield (g)(ns) (CV=38.45%)	Non-marketable yield (g)(ns) (CV=31.94)
19256.67	7854.00	3548.33
15281.67	6633.33	2481.67
13765.00	5903.33	2065.00

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, meaning conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

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Applying vermicast in the medium used for growing radish statistically gave the best growth performance in terms of width of the widest leaf. Also, result showed that vermicast is marginally better than ordinary compost than ordinary compost due to its higher nitrogen, phosphorus, and potassium content and ability to improve the soil structure and increase water-holding capacity (Singh, 2020), but study showed statistically it is not viable, or until proven. Further study should be conducted using these treatments on another crop.

(14) Output (Indicate the products, processes, or services the Project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

(17) Acknowledgments (Include only essential persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The researchers would like to acknowledge DA-BAR for the funds.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, hanging indention with 12-point space in between.)

Azarmi, R.; Giglou, M. T.; Taleshmikail, R. D. 2008. Influence of vermicompost on soil chemical and physical properties in tomato (*Lycopersicum esculentum*) field. African Journal of Biotechnology, Lagos, v. 7, n. 14, p. 2397-2401,2008.

Iderawumi, Abdulraheem. (2022). Re: Which would be better in-between "compost" and "vermicompost" for agricultural use?. Retrieved from: <https://www.researchgate.net/post/Which-would-be-better-in-between-compost-and-vermicompost-for-agricultural-use/62591e29d15888208574389f/citation/download>.

Joshi. R., J.Singh and A.P.Vig. 2014. Vermicompost is an effective organic fertilizer and biocontrol agent: effect on growth, yield, and quality of plants. Rev Environ Sci Biotechnol. DOI 10.1007/s11157-014-9347-1.

Lazcano, C.; Domínguez, J. 2011. The use of vermicompost in sustainable agriculture impacts plant growth and soil fertility. In: MIRANSARI, M. (Ed.). Soil nutrients. New York: Nova Science Publishers. chap. 10, p. 211-234.

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Lim, S. L., T. Y. Wu, P. N. Lim and K. P. Y. Shak. 2015. The use of vermicompost in organic farming: overview, effects on soil and economics. J Sci Food Agric. 95 (6): 1143-56.

National Action Plan (NAP). 2004. The Philippines National Action Plan to Ombatdesertication, Land Degradation, Drought and Poverty for 2004-2010. Department of Agriculture, Department of Environment and Natural Resources, Department of Science and Technology, and Department of Agrarian Reform. Manila, Philippines

Philippine Council for Agriculture, Forestry, Natural Resources, and Research Development. 2006. The Philippines Recommends Organic Fertilizer Production and Utilization. Los Banos, Laguna

Schippers RR. 2004. *Raphanus sativus* L. [online] Record from Protabase. Grubben GJH, Denton OA, editors. PROTA (Plant Resources of Tropical Africa / Ressources végétales de l'Afrique tropicale), Wageningen, Netherlands. Available from: <http://database.prota.org/search.htm>. Date accessed: 22 January 2010.

Singh, Arvind. (2020). Re: Which would be better in-between "compost" and "vermicompost" for agricultural use?. Retrieved from: <https://www.researchgate.net/post/Which-would-be-better-in-between-compost-and-vermicompost-for-agricultural-use/5fd1abd128d4932f1b4f13b3/citation/download.>)

<https://ap.fftc.org.tw/article/2952>

<https://aggie-horticulture.tamu.edu/food-technology/nutrition/radishes/>

<https://www.gov.nl.ca/ffa/files/agrifoods-plants-pdf-radish.pdf>